

PORTAGE COUNTY REGIONAL, OH, USA

WMO: 725253

Lat: 41.217N

Lon: 81.250W

Elev: 365

StdP: 97.01

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00380

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.2	-22.2	0.5	-15.8	-19.1	0.7	-12.9	10.7	-2.4	9.2	-3.8	2.3	250	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.9	21.6	29.5	21.0	28.1	20.2	23.6	27.7	22.8	26.9	22.0	26.2	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	25.2	21.6	17.0	24.6	20.9	16.2	24.1	72.7	27.5	69.2	27.5	66.2	26.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.3	7.3	6.2	-19.8	32.9	3.4	1.7	-22.2	34.1	-24.2	35.0	-26.1	36.0	-28.6	37.2
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.5	-2.0	2.7	9.5	16.0	19.7	22.0	21.0	17.6	11.4	4.9	0.4	(6)
	DBStd	10.16	6.45	6.75	6.15	5.11	4.75	3.20	2.86	2.67	3.98	4.75	4.95	5.24	(7)
	HDD10.0	1584	421	339	241	70	8	0	0	0	1	39	166	301	(8)
	HDD18.3	3377	678	569	484	269	101	22	4	8	62	221	403	557	(9)
	CDD10.0	1595	1	3	15	54	196	291	371	339	228	83	12	2	(10)
	CDD18.3	347	0	0	0	3	29	63	118	89	39	6	0	0	(11)
	CDH23.3	2705	0	0	4	42	273	487	942	631	293	33	0	0	(12)
	CDH26.7	676	0	0	0	4	51	113	282	144	78	4	0	0	(13)
(14) Wind	WSAvg	2.5	3.2	3.0	3.0	3.1	2.3	2.1	1.6	1.5	1.7	2.2	2.7	2.9	(14)
(15) Precipitation	PrecAvg	996	72	54	71	87	101	90	112	98	92	79	70	72	(15)
	PrecMax	1305	149	125	146	146	199	149	290	205	182	156	143	175	(16)
	PrecMin	651	35	26	13	41	40	13	31	27	31	27	36	23	(17)
	PrecStd	150	29	25	30	27	46	39	64	41	40	34	26	31	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	18.5	23.4	27.5	30.1	31.3	33.0	31.2	31.3	26.7	21.3	16.1	(19)
		MCWB	11.9	12.7	14.9	15.7	20.4	21.7	21.8	21.7	21.1	19.1	14.3	13.1	(20)
	2%	DB	12.4	14.1	18.8	24.3	27.9	29.5	31.1	29.8	29.0	23.9	17.5	13.6	(21)
		MCWB	10.6	10.4	13.0	14.9	18.9	20.4	21.8	21.3	21.1	17.4	11.9	10.8	(22)
	5%	DB	8.7	11.2	15.9	21.6	26.5	27.8	29.7	28.3	26.9	21.6	15.2	11.3	(23)
		MCWB	6.3	8.2	10.7	13.5	18.4	19.6	21.5	20.6	20.2	16.3	10.4	8.8	(24)
	10%	DB	6.0	7.3	12.8	18.8	24.6	26.2	28.0	27.1	24.6	19.4	12.9	8.2	(25)
		MCWB	4.3	4.1	8.3	12.3	17.0	18.6	20.4	20.2	19.2	15.5	9.0	6.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	14.0	16.2	17.5	22.0	23.2	25.0	24.1	24.2	21.1	15.5	13.9	(27)
		MCDB	13.9	17.9	21.0	23.5	27.9	28.9	29.1	27.5	27.6	25.0	19.5	15.4	(28)
	2%	WB	10.7	11.7	13.8	16.1	20.6	22.1	24.0	23.0	22.8	19.4	13.6	11.6	(29)
		MCDB	11.7	13.1	17.8	22.4	25.5	27.0	28.2	26.9	26.5	22.0	16.0	12.9	(30)
	5%	WB	7.1	7.8	11.8	14.9	19.6	21.2	23.0	22.3	21.6	17.7	11.3	8.5	(31)
		MCDB	8.1	11.3	15.4	20.1	24.7	25.5	27.1	26.4	25.4	20.3	14.6	10.3	(32)
	10%	WB	3.5	4.3	9.1	13.2	18.5	20.4	22.2	21.4	20.2	15.8	9.2	5.8	(33)
		MCDB	5.7	7.1	12.6	17.9	23.2	24.4	26.0	25.4	23.4	19.1	12.7	8.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.6	9.3	11.3	11.3	10.6	10.7	10.7	10.7	9.7	8.5	6.5	(35)
		MCDBR	10.6	12.2	13.2	15.0	13.2	12.4	12.0	12.3	12.6	12.8	12.5	10.4	(36)
	5% WB	MCWBR	9.2	9.2	8.1	8.2	6.3	5.9	5.0	5.3	5.7	7.1	8.4	9.2	(37)
		MCWBR	9.3	11.4	12.2	13.1	11.6	10.6	9.9	9.8	10.5	10.4	10.7	10.0	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.327	0.355	0.386	0.424	0.445	0.448	0.436	0.383	0.359	0.317	0.300	(40)
		taud	2.433	2.329	2.335	2.292	2.206	2.187	2.195	2.222	2.345	2.410	2.504	2.480	(41)
	Ebn at Noon	Ebn at Noon	858	877	892	885	855	835	828	827	851	832	831	827	(42)
		Edh at Noon	78	101	113	127	142	145	142	135	111	93	73	70	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.28	3.26	4.45	5.33	5.86	5.87	5.25	4.21	2.71	1.74	1.14	(44)
	RadStd	RadStd	0.13	0.28	0.33	0.42	0.46	0.46	0.41	0.27	0.39	0.25	0.27	0.11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page